Topoi (ILF)

Goldblatt / Exercises / Reading group

Summer 2026

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Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n, consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \ldots \leq n-1$. How many arrows does $\mathbf{n}$ have?

> **Solution**
>
> For each natural number $k < n$, we can count the number of incoming arrows as
>
> $$|\{j \in \{0, 1, \ldots, n-1\} : j \leq k\}| = k + 1.$$
>
> Therefore, there are
>
> $$\sum_{k=0}^{n-1}(k+1) = \binom{n}{2}$$
>
> arrows in total.

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathscr{C}$ are valid subcategories?

- The empty category $\mathscr{C} = \mathbf{0}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \ldots\}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \ldots\}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \ldots\}$

> **Solution**
>
> - The empty category $\mathscr{C} = \mathbf{0}$
>
> The two conditions for being a subcategory are vacuously true; in fact, the empty category is a

subcategory of every other category. Let $\mathscr{D}$ be any category. Then:

(i) Every $\mathbf{0}$-object is a $\mathscr{D}$-object, vacuously.

(ii) For every pair of $\mathbf{0}$-objects a, b, the inclusion $\mathbf{0}(a, b) \subseteq \mathscr{D}(a, b)$ holds, vacuously.

- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{0\} = \{1, 2, 3, 4, ...\}$

Subcategories must be closed under identities. Since the identity is omitted from the given subquiver, it is not a subcategory.

- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{1\} = \{0, 2, 3, 4, ...\}$

The given data already form a subquiver. It suffices to check that the subquiver is a category itself with the inherited identity and composition operations. This amounts to checking that the set $N \setminus \{1\}$ is a submonoid of the natural numbers.

1. Since $0 \in N \setminus \{1\}$, it is closed under identities.
2. Let $x, y \in N \setminus \{1\}$. If either $x = 0$ or $y = 0$, then $x + y \in \{x, y\} \subseteq N \setminus \{1\}$. The remaining case is when neither are 0, in which case $x, y \geq 2$. Therefore, $x + y \geq 4$ and so $x + y \in N \setminus \{1\}$.

- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{2\} = \{0, 1, 3, 4, ...\}$

The given subquiver is not closed under composition: $1 \in N \setminus \{2\}$ yet $1 + 1 \notin N \setminus \{2\}$. Therefore, the given subquiver is not a subcategory.

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathscr{C}$ be a category and X be an object of $\mathscr{C}$. Verify that the collection $\mathscr{C}(X, X)$ of all $\mathscr{C}$-arrows from X to X forms a monoid with respect to composition.

Solution

We shall check that $(\mathscr{C}(X, X), \circ|_{\mathscr{C}(X,X) \times \mathscr{C}(X,X)}, 1_X)$ is a monoid.

Notes

It is worth mentioning that the "hom-collection" $\mathscr{C}(X, X)$ need not be a "small" set whenever $\mathscr{C}$ is not a "locally small" category. In that case, we would call $\mathscr{C}(X, X)$ a "large" (i.e. proper class-sized) monoid.

Let $\mathscr{C}_1$ denote the collection of arrows of $\mathscr{C}$. Notice that $\mathscr{C}(X, X) \times \mathscr{C}(X, X) \subseteq \mathscr{C}_1 {}_{\mathrm{cod}}\!\times_{\mathrm{dom}} \mathscr{C}_1$

and that $\circ(\mathscr{C}(X,X) \times \mathscr{C}(X,X)) \subseteq \mathscr{C}(X,X)$. Therefore, $\circ$ restricts to a binary operation on the collection $\mathscr{C}(X,X)$.

Notes

It is also worth mentioning that Goldblatt defines composition as an operation with typing $\circ : \mathscr{C}_1 {}_{\mathrm{dom}}\!\times_{\mathrm{cod}} \mathscr{C}_1 \to \mathscr{C}_1$, whereas we are using the typing $\circ : \mathscr{C}_1 {}_{\mathrm{cod}}\!\times_{\mathrm{dom}} \mathscr{C}_1 \to \mathscr{C}_1$. The two operations are essentially the same, up to swapping arguments.

The associativity of composition in a category translates to associativity of composition on subset $\mathscr{C}(X,X)$. Pedantically, for $f, g, h \in \mathscr{C}(X,X)$:

$$\begin{aligned}
&h \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} (g \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} f) \\
&= h \circ (g \circ f) \\
&= (h \circ g) \circ f \\
&= (h \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} g) \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} f
\end{aligned}$$

Likewise, $1_X \in \mathscr{C}(X,X)$ acts as an identity for compositions from $\mathscr{C}_1$ that are defined. Therefore, 1_X also acts as an identity on the subset $\mathscr{C}(X,X)$, for which composition is always defined. Pedantically, for $f \in \mathscr{C}(X,X)$:

$$\begin{aligned}
1_X \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} f &= 1_X \circ f = f \\
f \circ|_{\mathscr{C}(X,X)\times\mathscr{C}(X,X)} 1_X &= f \circ 1_X = f
\end{aligned}$$

Therefore, $(\mathscr{C}(X,X), \circ, 1_X)$ is a monoid (modulo the fact that the collection $\mathscr{C}(X,X)$ can be a proper class).

1.2.2 Sum Category

Let $\mathscr{C}$ and $\mathscr{D}$ be categories. The sum category, denoted $\mathscr{C}+\mathscr{D}$, is a category whose collection of objects is the disjoint union of $\mathscr{C}$-objects and $\mathscr{D}$-objects, and whose collection of arrows is the disjoint union of $\mathscr{C}$-arrows and $\mathscr{D}$-arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathscr{C}$ and $\mathscr{D}$. Verify that $\mathscr{C} + \mathscr{D}$ is a category.
- Verify that there are no $(\mathscr{C} + \mathscr{D})$-arrows between any object that came from $\mathscr{C}$ and any object that came from $\mathscr{D}$.
- Find an example of $\mathscr{C}$ and $\mathscr{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

> **Notes**
> For collections A and B, let $A + B$ denote the disjoint union. To be pedantic:
>
> $$A + B = \{\langle a, 0\rangle : a \in A\} \cup \{\langle b, 1\rangle : b \in B\}$$
>
> If either A or B is a “large” set (i.e. a proper class), then $A + B$ is also a “large” set. Also note that $A + B$ is “small” whenever A and B are both “small”.
> For operations $f : A \to C$ and $g : B \to D$ between collections, let $f + g : A + B \to C + D$ denote the operation defined as follows:
>
> $$\begin{aligned}(f+g)(\langle a, 0\rangle) &= \langle f(a), 0\rangle \\ (f+g)(\langle b, 1\rangle) &= \langle f(b), 1\rangle\end{aligned}$$

Solution Part I

Let $\mathscr{C} = (O, A, \mathrm{dom}, \mathrm{cod}, \circ, 1)$ and $\mathscr{D} = (O', A', \mathrm{dom}', \mathrm{cod}', \circ', 1')$ be the two categories. The typing of our operations is as follows:

$$\begin{aligned}\mathrm{dom} &: A \to O & \mathrm{dom}' &: A' \to O' \\ \mathrm{cod} &: A \to O & \mathrm{cod}' &: A' \to O' \\ \circ &: A \;{}_{\mathrm{dom}}{\times}_{\mathrm{cod}}\; A \to A & \circ' &: A' \;{}_{\mathrm{dom}'}{\times}_{\mathrm{cod}'}\; A' \to A' \\ 1 &: O \to A & 1' &: O' \to A'\end{aligned}$$

(Here, we choose the typing of $\circ$ and $\circ'$ to follow Goldblatt’s convention: $\circ(\langle g, f\rangle) = g \circ f$ whenever $f : a \to b$ and $g : b \to c$.)

As a first approximation, we wish to claim that $\mathscr{C} + \mathscr{D} = (O + O', A + A', \mathrm{dom} + \mathrm{dom}', \mathrm{cod} + \mathrm{cod}', \circ + \circ', 1 + 1')$ is a valid category. The proposed operations would have the following typing:

$$\begin{aligned}\mathrm{dom} + \mathrm{dom}' &: A + A' \to O + O' \\ \mathrm{cod} + \mathrm{cod}' &: A + A' \to O + O' \\ \circ + \circ' &: (A \;{}_{\mathrm{dom}}{\times}_{\mathrm{cod}}\; A) + (A' \;{}_{\mathrm{dom}}{\times}_{\mathrm{cod}}\; A') \to A + A' \\ 1 + 1' &: O + O' \to A + A'\end{aligned}$$

This is close to what we need, but our proposed composition does not have the required typing. To be pedantic, we can adjust the domain of $\circ + \circ'$ by precomposing with the following (bijective) operation:

$$(A + A') \;{}_{\mathrm{dom}+\mathrm{dom}'}{\times}_{\mathrm{cod}+\mathrm{cod}'}\; (A + A') \xrightarrow{\phi} (A \;{}_{\mathrm{dom}}{\times}_{\mathrm{cod}}\; A) + (A' \;{}_{\mathrm{dom}}{\times}_{\mathrm{cod}}\; A')$$

$$\langle\langle g, 0\rangle, \langle f, 0\rangle\rangle \longmapsto \langle\langle g, f\rangle, 0\rangle \qquad \text{for } a \xrightarrow{f} b \xrightarrow{g} c \text{ in } \mathscr{C}$$

$$\langle\langle g', 1\rangle, \langle f', 1\rangle\rangle \longmapsto \langle\langle g', f'\rangle, 1\rangle \qquad \text{for } a' \xrightarrow{f'} b' \xrightarrow{g'} c' \text{ in } \mathscr{D}$$

To verify that we have specified all possible input combinations in our definition of ϕ, note that for each $h \in A + A'$ and $n \in \{0, 1\}$, we have that $(\mathrm{dom} + \mathrm{dom}')(\langle h, n\rangle) = n = (\mathrm{cod} + \mathrm{cod}')(\langle h, n\rangle)$.

Therefore:

$$\begin{aligned}
&(A + A')\ {}_{\mathrm{dom}+\mathrm{dom}'}{\times}_{\mathrm{cod}+\mathrm{cod}'}\ (A + A') \\
&= \big\{\langle\langle h, m\rangle, \langle k, n\rangle\rangle \in (A + A')^2 : (\mathrm{dom}+\mathrm{dom}')(\langle h, m\rangle) = (\mathrm{cod}+\mathrm{cod}')(\langle k, n\rangle)\big\} \\
&= \big\{\langle\langle h, m\rangle, \langle k, n\rangle\rangle \in (A + A')^2 : m = n\big\}
\end{aligned}$$

Now, we claim that $\mathscr{C} + \mathscr{D} = (O + O', A + A', \mathrm{dom}+\mathrm{dom}', \mathrm{cod}+\mathrm{cod}', (\circ + \circ')\phi, 1 + 1')$ satisfies the axioms of a category. We do so as follows:

- Domain of Composition:

$$\begin{aligned}
&(\mathrm{dom}+\mathrm{dom}')\big((\circ + \circ')\phi(\langle\langle g, 0\rangle, \langle f, 0\rangle\rangle)\big) \\
&= (\mathrm{dom}+\mathrm{dom}')\big((\circ + \circ')(\langle\langle g, f\rangle, 0\rangle)\big) \\
&= (\mathrm{dom}+\mathrm{dom}')(\langle \circ(\langle g, f\rangle), 0\rangle) \\
&= \langle \mathrm{dom}(\circ(\langle g, f\rangle)), 0\rangle \\
&= \langle \mathrm{dom}(g \circ f), 0\rangle \\
&= \langle \mathrm{dom}(f), 0\rangle \\
&= (\mathrm{dom}+\mathrm{dom}')(\langle f, 0\rangle)
\end{aligned}$$

  and by symmetry (with computation omitted):

$$(\mathrm{dom}+\mathrm{dom}')\big((\circ + \circ')\phi\big(\langle\langle g', 1\rangle, \langle f', 1\rangle\rangle\big)\big) = (\mathrm{dom}+\mathrm{dom}')(\langle f', 1\rangle)$$

- Codomain of Composition:

  Computations follow by a symmetric argument to verifying domain of composition. With computations omitted, we have the following:

$$(\mathrm{cod}+\mathrm{cod}')\big((\circ + \circ')\phi(\langle\langle g, 0\rangle, \langle f, 0\rangle\rangle)\big) = (\mathrm{cod}+\mathrm{cod}')(\langle g, 0\rangle)$$

  and

$$(\mathrm{cod}+\mathrm{cod}')\big((\circ + \circ')\phi\big(\langle\langle g', 1\rangle, \langle f', 1\rangle\rangle\big)\big) = (\mathrm{cod}+\mathrm{cod}')(\langle g', 1\rangle)$$

- Associativity:

$$\begin{aligned}
&(\circ + \circ')\phi\big(\langle(\circ + \circ')\phi(\langle\langle h, 0\rangle, \langle g, 0\rangle\rangle), \langle f, 0\rangle\rangle\big) \\
&= (\circ + \circ')\phi\big(\langle(\circ + \circ')(\langle\langle h, g\rangle, 0\rangle), \langle f, 0\rangle\rangle\big) \\
&= (\circ + \circ')\phi(\langle\langle \circ(\langle h, g\rangle), 0\rangle, \langle f, 0\rangle\rangle) \\
&= (\circ + \circ')(\langle\langle \circ(\langle h, g\rangle), f\rangle, 0\rangle) \\
&= \langle \circ(\langle \circ(\langle h, g\rangle), f\rangle), 0\rangle \\
&= \langle \circ(\langle h \circ g, f\rangle), 0\rangle \\
&= \langle (h \circ g) \circ f, 0\rangle
\end{aligned}$$

  and by symmetry (with computation omitted):

$$(\circ + \circ')\phi\big(\langle\langle h, 0\rangle, (\circ + \circ')\phi(\langle\langle g, 0\rangle, \langle f, 0\rangle\rangle)\rangle\big) = \langle h \circ (g \circ f), 0\rangle$$

Since $(h \circ g) \circ f = h \circ (g \circ f)$ holds in $\mathscr{C}$, it follows that

$$\begin{aligned}
&(\circ + \circ')\phi\big(\langle(\circ + \circ')\phi(\langle\langle h, 0\rangle, \langle g, 0\rangle\rangle), \langle f, 0\rangle\rangle\big) \\
&= \langle (h \circ g) \circ f, 0\rangle \\
&= \langle h \circ (g \circ f), 0\rangle \\
&= (\circ + \circ')\phi\big(\langle\langle h, 0\rangle, (\circ + \circ')\phi(\langle\langle g, 0\rangle, \langle f, 0\rangle\rangle)\rangle\big)
\end{aligned}$$

and symmetrically (with computation omitted) for

$$\begin{aligned}
&(\circ + \circ')\phi\big(\langle(\circ + \circ')\phi(\langle\langle h', 1\rangle, \langle g', 1\rangle\rangle), \langle f', 1\rangle\rangle\big) \\
&= \langle (h' \circ g') \circ f', 1\rangle \\
&= \langle h' \circ (g' \circ f'), 1\rangle \\
&= (\circ + \circ')\phi\big(\langle\langle h', 1\rangle, (\circ + \circ')\phi(\langle\langle g', 1\rangle, \langle f', 1\rangle\rangle)\rangle\big)
\end{aligned}$$

- Domain of identity:

$$\begin{aligned}
&(\mathrm{dom} + \mathrm{dom}')\big((\mathsf{1} + \mathsf{1}')(\langle a, 0\rangle)\big) \\
&= (\mathrm{dom} + \mathrm{dom}')(\langle \mathsf{1}(a), 0\rangle) \\
&= \langle \mathrm{dom}(\mathsf{1}(a)), 0\rangle \\
&= \langle a, 0\rangle
\end{aligned}
\qquad
\begin{aligned}
&(\mathrm{dom} + \mathrm{dom}')\big((\mathsf{1} + \mathsf{1}')(\langle a', 1\rangle)\big) \\
&= (\mathrm{dom} + \mathrm{dom}')(\langle \mathsf{1}'(a'), 1\rangle) \\
&= \langle \mathrm{dom}'(\mathsf{1}'(a')), 1\rangle \\
&= \langle a', 1\rangle
\end{aligned}$$

- Codomain of identity:

$$\begin{aligned}
&(\mathrm{cod} + \mathrm{cod}')\big((\mathsf{1} + \mathsf{1}')(\langle a, 0\rangle)\big) \\
&= (\mathrm{cod} + \mathrm{cod}')(\langle \mathsf{1}(a), 0\rangle) \\
&= \langle \mathrm{cod}(\mathsf{1}(a)), 0\rangle \\
&= \langle a, 0\rangle
\end{aligned}
\qquad
\begin{aligned}
&(\mathrm{cod} + \mathrm{cod}')\big((\mathsf{1} + \mathsf{1}')(\langle a', 1\rangle)\big) \\
&= (\mathrm{cod} + \mathrm{cod}')(\langle \mathsf{1}'(a'), 1\rangle) \\
&= \langle \mathrm{cod}'(\mathsf{1}'(a')), 1\rangle \\
&= \langle a', 1\rangle
\end{aligned}$$

- Right identity:

$$\begin{aligned}
&(\circ + \circ')\phi\big(\langle\langle f, 0\rangle, (\mathsf{1} + \mathsf{1}')\big((\mathrm{dom} + \mathrm{dom}')(\langle f, 0\rangle)\big)\rangle\big) \\
&= (\circ + \circ')\phi\big(\langle\langle f, 0\rangle, (\mathsf{1} + \mathsf{1}')(\langle \mathrm{dom}(f), 0\rangle)\rangle\big) \\
&= (\circ + \circ')\phi(\langle\langle f, 0\rangle, \langle \mathsf{1}(\mathrm{dom}(f)), 0\rangle\rangle) \\
&= (\circ + \circ')(\langle\langle f, \mathsf{1}(\mathrm{dom}(f))\rangle, 0\rangle) \\
&= \langle \circ(\langle f, \mathsf{1}(\mathrm{dom}(f))\rangle), 0\rangle \\
&= \langle f, 0\rangle
\end{aligned}$$

and symmetrically (with computation omitted) for

$$(\circ + \circ')\phi\big(\langle\langle f', 1\rangle, (\mathsf{1} + \mathsf{1}')\big((\mathrm{dom} + \mathrm{dom}')(\langle f', 1\rangle)\big)\rangle\big) = \langle f', 1\rangle$$

- Left identity:

  Computations follow by a symmetric argument to verifying right identity. With computations omitted, we have the following:

  $$(\circ + \circ')\phi\big(\langle(\mathsf{1} + \mathsf{1}')\big((\mathrm{dom} + \mathrm{dom}')(\langle f, 0\rangle)\big), \langle f, 0\rangle\rangle\big) = \langle f, 0\rangle$$

  and

  $$(\circ + \circ')\phi\big(\langle(\mathsf{1} + \mathsf{1}')\big((\mathrm{dom} + \mathrm{dom}')\big(\langle f', 1\rangle\big)\big), \langle f', 1\rangle\rangle\big) = \langle f', 1\rangle$$

Solution Part II

This follows from the construction above. If $\langle h, m\rangle$ is composable with $\langle k, n\rangle$, then $m = n$ follows, as shown above.

Solution Part III

Consider the categories $(\{0\}, \{1\}, !, !, !, !)$ and $(\{0\}, \{2\}, !, !, !, !)$ where ! is a placeholder for the unique possible function in that slot. Had we used set union instead of disjoint union, we would have obtained a quiver with one vertex 0 and two edges 1 and 2. Although we would have $1 \circ 1 = 1$ and $2 \circ 2 = 2$, it is unclear how to define $1 \circ 2$ and $2 \circ 1$. In particular, the ϕ used in the construction above would no longer be defined on all inputs.

1.3 Extra Exercise

1.3.1 Quivers

A *quiver* is a directed graph that allows loops and multiple parallel edges. In particular, a quiver Q comprises:

- a collection of things called Q-*vertices*;
- a collection of things called Q-*edges*;
- operations assigning to each Q-edge f a Q-vertex $s(f)$ (the "source" of f) and a Q-vertex $t(f)$ (the "target" of f).

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \mathrm{dom}(f)$, and $t(f) = \mathrm{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.

> **Solution**
>
> There are many possible solutions to this exercise. Here, we'll cover the smallest solution:

> Let $M = (\mathbb{Z}/2\mathbb{Z}, +, 0)$ and $N = (\mathbb{Z}/2\mathbb{Z}, \cdot, 1)$ be the additive monoid and multiplicative monoid of $\mathbb{Z}/2\mathbb{Z}$, respectively. The categories associated to these monoids each have one object and two arrows. Therefore, their underlying quivers are isomorphic.
> The second category satisfies the first-order formula $(\forall f)(f \circ f = f)$ as f varies over all arrows, whereas the first category does not. Therefore, the two categories cannot be isomorphic.

Meeting III

Category Theory Pt. 2

(work in progress)

2.1 Minimal Exercises

2.1.1 Iso Arrows

Exercises from Goldblatt section 3.3: in any category, verify the following:

1. Every identity arrow is iso.

> **Solution**
>
> Let 1_x be an identity arrow for an object x. By the identity axioms of a category, $1_x \circ 1_x = 1_x$ and $1_x \circ 1_x = 1_x$. Therfore, 1_x is an isomorphism with inverse 1_x.

2. If $b \xrightarrow{f} c$ is iso, so is $c \xrightarrow{f^{-1}} b$.

> **Solution**
>
> By definition of iso, $f^{-1} \circ f = 1_b$ and $f \circ f^{-1} = 1_c$. Therefore (upon swapping the order of conjuncts), $f \circ f^{-1} = 1_c$ and $f^{-1} \circ f = 1_b$. Thus, f^{-1} is an iso (with inverse f).

3. $a \xrightarrow{f \circ g} c$ is iso if $b \xrightarrow{f} c$, $a \xrightarrow{g} b$ are, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

> **Solution**
>
> By computation:
>
> $$\begin{aligned} &\left(g^{-1} \circ f^{-1}\right) \circ (f \circ g) \\ &= g^{-1} \circ \left(f^{-1} \circ (f \circ g)\right) \\ &= g^{-1} \circ \left(\left(f^{-1} \circ f\right) \circ g\right) \\ &= g^{-1} \circ (1_b \circ g) \\ &= g^{-1} \circ g \\ &= 1_a \end{aligned} \qquad \begin{aligned} &(f \circ g) \circ \left(g^{-1} \circ f^{-1}\right) \\ &= f \circ \left(g \circ \left(g^{-1} \circ f^{-1}\right)\right) \\ &= f \circ \left(\left(g \circ g^{-1}\right) \circ f^{-1}\right) \\ &= f \circ \left(1_b \circ f^{-1}\right) \\ &= f \circ f^{-1} \\ &= 1_c \end{aligned}$$

2.1.2 Terminal Objects

Selected exercises from Goldblatt section 3.6:

1. For any category $\mathscr{C}$, prove that all terminal $\mathscr{C}$-objects (if any) are isomorphic (to each other).

> **Solution**
>
> Let T and U be terminal objects. Since T is terminal, there is an arrow $\mathsf{l}_U : U \to T$. Since U is terminal, there is an arrow $\mathsf{l}'_T : T \to U$. Therefore, we have the following composites:
>
> $$\begin{aligned} \mathsf{l}_U \circ \mathsf{l}'_T &: T \to T \\ \mathsf{l}'_T \circ \mathsf{l}_U &: U \to U \end{aligned}$$
>
> Since T is terminal, parallel arrows into T coincide. Therefore, $\mathsf{l}_U \circ \mathsf{l}'_T = \mathsf{1}_T : T \to T$. Since U is terminal, parallel arrows into U coincide. Therefore, $\mathsf{l}'_T \circ \mathsf{l}_U = \mathsf{1}_U : U \to U$. Thus, l_U and l'_T are each iso (and inverse to each other). Therefore, $T \cong U$.

2. Find terminals in $\mathbf{Set}^2$, $\mathbf{Set}^{\to}$, and the poset $\mathbf{n}$.

> **Solution**
>
> - A terminal in $\mathbf{Set}^2$ is $\langle 1, 1 \rangle$. Given any other object $\langle X, Y \rangle$, an arrow is $\langle \mathsf{l}_X, \mathsf{l}_Y \rangle$. If $\langle f, g \rangle : \langle X, Y \rangle \to \langle 1, 1 \rangle$ is an arrow, then $f : X \to 1$ and $g : Y \to 1$ as arrows in $\mathbf{Set}$. Therefore $f = \mathsf{l}_X$ and $g = \mathsf{l}_Y$ since parallel arrows into terminals coincide.
> - A terminal in $\mathbf{Set}^{\to}$ is $1 \xrightarrow{\mathsf{1}_1} 1$. Given any other object $f : X \to Y$, an arrow is $\langle \mathsf{l}_X, \mathsf{l}_Y \rangle$ since the following square commutes in **Set** (being parallel arrows into a terminal):
>
> $$\begin{tikzcd} X \arrow[r, "\mathsf{l}_X"] \arrow[d, "f"'] & 1 \arrow[d, "\mathsf{1}_1"] \\ Y \arrow[r, "\mathsf{l}_Y"] & 1 \end{tikzcd}$$
>
>   If $\langle f, g \rangle : \langle X, Y \rangle \to \langle 1, 1 \rangle$ is an arrow, then $f : X \to 1$ and $g : Y \to 1$ as arrows in $\mathbf{Set}$. Therefore $f = \mathsf{l}_X$ and $g = \mathsf{l}_Y$ since parallel arrows into terminals coincide.
> - The terminal in the poset $\mathbf{n}$ is $n-1$. Note that this is only possible when $n > 0$, because the empty category cannot have a terminal object.
>
>   If x is any object, then $x \leq n - 1$ since $n - 1$ is the maximum element. Therefore, there exists an arrow into $n - 1$ from any other object. Since all parallel pairs coincide in a poset, such an arrow $x \to n - 1$ is unique.

2.1.3 Products

Selected exercises from Goldblatt section 3.8:

Let a, b be two $\mathscr{C}$-objects that have a product $a \times b$ with projections $\mathrm{pr}_a : a \times b \to a$ and $\mathrm{pr}_b : a \times b \to b$.

Exercise 1: Prove that $\langle \mathrm{pr}_a, \mathrm{pr}_b \rangle = 1_{a \times b}$.

Solution

The following diagrams are commutative:

$$\begin{tikzcd}
& a \times b \arrow[dl, "\mathrm{pr}_a"'] \arrow[d, "\langle \mathrm{pr}_a, \mathrm{pr}_b \rangle"] \arrow[dr, "\mathrm{pr}_b"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b
\end{tikzcd}
\qquad
\begin{tikzcd}
& a \times b \arrow[dl, "\mathrm{pr}_a"'] \arrow[d, "1_{a \times b}"] \arrow[dr, "\mathrm{pr}_b"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b
\end{tikzcd}$$

The first diagram commutes by definition of the comparison morphism $\langle \mathrm{pr}_a, \mathrm{pr}_b \rangle$. The second diagram commutes by the right identity axiom.
By uniqueness of cone morphisms into a product, it follows that $\langle \mathrm{pr}_a, \mathrm{pr}_b \rangle = 1_{a \times b}$.

Exercise 2: Prove that if $\langle f, g \rangle = \langle k, h \rangle$, then $f = k$ and $g = h$.

Solution

By definition of the comparison morphism into a product cone, the following diagrams are commutative:

$$\begin{tikzcd}
& x \arrow[dl, "f"'] \arrow[d, "\langle f, g \rangle"] \arrow[dr, "g"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b
\end{tikzcd}
\qquad
\begin{tikzcd}
& x \arrow[dl, "k"'] \arrow[d, "\langle k, h \rangle"] \arrow[dr, "h"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b
\end{tikzcd}$$

Therefore, $f = \mathrm{pr}_a \circ \langle f, g \rangle = \mathrm{pr}_a \circ \langle k, h \rangle = k$ and $h = \mathrm{pr}_b \circ \langle f, g \rangle = \mathrm{pr}_b \circ \langle k, h \rangle = h$

Exercise 3: Prove that $\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h$.

Solution

> Consider the following diagram:
>
> $$\begin{tikzcd}
> & & x \arrow[dl, "h"'] \arrow[d, "h"] \arrow[dr, "h"] & & \\
> & y \arrow[dl, "f"'] & y \arrow[d, "\langle f,g \rangle"] & y \arrow[dr, "g"] & \\
> a & & a \times b \arrow[ll, "\mathrm{pr}_a"] \arrow[rr, "\mathrm{pr}_b"'] & & b
> \end{tikzcd}$$
>
> This diagram commutes because
>
> $$\begin{aligned}
> \mathrm{pr}_a \circ (\langle f, g\rangle \circ h) = (\mathrm{pr}_a \circ \langle f, g\rangle) \circ h = f \circ h \\
> \mathrm{pr}_b \circ (\langle f, g\rangle \circ h) = (\mathrm{pr}_b \circ \langle f, g\rangle) \circ h = g \circ h
> \end{aligned}$$
>
> By uniquness of the comparison arrow, it follows that $\langle f \circ h, g \circ h\rangle = \langle f, g\rangle \circ h$.

Exercise 4: Show that if $\mathscr{C}$ has a terminal object and products, then for any $\mathscr{C}$-object a, $a \cong a \times 1$ and indeed $\langle 1_a, \mathsf{l}_a\rangle$ is iso.

> **Solution**
>
> Firstly, note by definition of the comparison morphism that $\mathrm{pr}_a \circ \langle 1_a, \mathsf{l}_a\rangle = 1_a$. Secondly, we utilize the previous exercise to compute that:
>
> $$\langle 1_a, \mathsf{l}_a\rangle \circ \mathrm{pr}_a = \langle 1_a \circ \mathrm{pr}_a, \mathsf{l}_a \circ \mathrm{pr}_a\rangle = \langle \mathrm{pr}_a, \mathsf{l}_a \circ \mathrm{pr}_a\rangle$$
>
> Notice the following arrows are parallel into a terminal: $\mathsf{l}_a \circ \mathrm{pr}_a : a \times 1 \to 1$ and $\mathrm{pr}_1 : a \times 1 \to 1$. It follows that $\mathsf{l}_a \circ \mathrm{pr}_a = \mathrm{pr}_1$. Combined with exercise 1, we have:
>
> $$\langle 1_a, \mathsf{l}_a\rangle \circ \mathrm{pr}_a = \langle 1_a \circ \mathrm{pr}_a, \mathsf{l}_a \circ \mathrm{pr}_a\rangle = \langle \mathrm{pr}_a, \mathsf{l}_a \circ \mathrm{pr}_a\rangle = \langle \mathrm{pr}_a, \mathrm{pr}_1\rangle = 1_{a \times 1}$$
>
> It follows that $\langle 1_a, \mathsf{l}_a\rangle$ is inverse to pr_a. Therefore, $a \cong a \times 1$ via the iso $\langle 1_a, \mathsf{l}_a\rangle$.

Exercise 5: Prove that $1_a \times 1_b = 1_{a \times b}$.

> **Solution**

The following diagram commutes by the left/right identity axioms:

$$\begin{tikzcd} a \arrow[d, "1_a"'] & a \times b \arrow[l, "\mathrm{pr}_a"'] \arrow[r, "\mathrm{pr}_b"] \arrow[d, "1_{a\times b}"] & b \arrow[d, "1_b"] \\ a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b \end{tikzcd}$$

Therefore, the middle arrow equals the comparison morphism: $1_{a\times b} = 1_a \times 1_b$.

Exercise 6: Prove that a product $b \times a$ exists and that $a \times b \cong b \times a$.

Solution

Part 1:

Suppose "the" product of a and b exists and that we have a product diagram:

$$a \xleftarrow{\mathrm{pr}_a} a \times b \xrightarrow{\mathrm{pr}_b} b$$

We claim that the following is a product diagram, thereby demonstrating the existence of $b \times a$:

$$b \xleftarrow{\mathrm{pr}_b} a \times b \xrightarrow{\mathrm{pr}_a} a$$

Let $f : x \to b$ and $g : x \to a$ be arbitrary arrows. Then, $\langle g, f\rangle : x \to a \times b$ is an arrow such that $\mathrm{pr}_b \circ \langle g, f\rangle = f$ and $\mathrm{pr}_a \circ \langle g, f\rangle = g$. This shows the existence aspect of the universal property for $b \times a$.

Suppose we have another arrow $h : x \to a \times b$ with the property that $\mathrm{pr}_b \circ h = g$ and $\mathrm{pr}_a \circ h = f$.

Then (by swapping order of conjuncts), $\mathrm{pr}_a \circ h = f$ and $\mathrm{pr}_b \circ h = g$.

It follows that $h = \langle g, f\rangle$ by utilizing the uniqueness aspect of the universal property of $a \times b$.

This shows the uniqueness aspect of the universal property for $b \times a$.

Therefore, $b \times a$ exists with the proposed product diagram.

Part 2:

Suppose that we have any other product diagram

$$b \xleftarrow{\mathrm{pr}'_b} b \times a \xrightarrow{\mathrm{pr}'_a} a$$

We would like to show that $b \times a \cong a \times b$ in a way that is compatible with the product diagrams.

Consider the following comparison arrows:

$$
\begin{tikzcd}
& b \times a \arrow[dl, "\mathrm{pr}'_a"'] \arrow[d, "\langle \mathrm{pr}'_a{,}\mathrm{pr}'_b \rangle"] \arrow[dr, "\mathrm{pr}'_b"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] & b
\end{tikzcd}
\qquad
\begin{tikzcd}
& a \times b \arrow[dl, "\mathrm{pr}_b"'] \arrow[d, "\langle \mathrm{pr}_b{,}\mathrm{pr}_a \rangle'"] \arrow[dr, "\mathrm{pr}_a"] & \\
b & b \times a \arrow[l, "\mathrm{pr}'_b"] \arrow[r, "\mathrm{pr}'_a"'] & a
\end{tikzcd}
$$

By "pasting" together the above commutative diagrams in a compatible manner (e.g. by horizontally flipping one of them and placing it below the other one), we have the following commutative diagrams:

$$
\begin{tikzcd}
& b \times a \arrow[dl, "\mathrm{pr}'_a"'] \arrow[d, "\langle \mathrm{pr}'_a{,}\mathrm{pr}'_b \rangle"] \arrow[dr, "\mathrm{pr}'_b"] & \\
a & a \times b \arrow[l, "\mathrm{pr}_a"] \arrow[r, "\mathrm{pr}_b"'] \arrow[d, "\langle \mathrm{pr}_b{,}\mathrm{pr}_a \rangle'"] & b \\
& b \times a \arrow[ul, "\mathrm{pr}'_a"] \arrow[ur, "\mathrm{pr}'_b"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
& a \times b \arrow[dl, "\mathrm{pr}_b"'] \arrow[d, "\langle \mathrm{pr}_b{,}\mathrm{pr}_a \rangle'"] \arrow[dr, "\mathrm{pr}_a"] & \\
b & b \times a \arrow[l, "\mathrm{pr}'_b"] \arrow[r, "\mathrm{pr}'_a"'] \arrow[d, "\langle \mathrm{pr}'_a{,}\mathrm{pr}'_b \rangle"] & a \\
& a \times b \arrow[ul, "\mathrm{pr}_b"] \arrow[ur, "\mathrm{pr}_a"'] &
\end{tikzcd}
$$

From the uniqueness aspect of the universal property of $b \times a$, the left diagram demonstrates that:

$$\langle \mathrm{pr}_b, \mathrm{pr}_a \rangle' \circ \langle \mathrm{pr}'_a, \mathrm{pr}'_b \rangle = 1_{b \times a}$$

Similarly, from the uniqueness aspect of the universal property of $a \times b$, the right diagram demonstrates that:

$$\langle \mathrm{pr}'_a, \mathrm{pr}'_b \rangle \circ \langle \mathrm{pr}_b, \mathrm{pr}_a \rangle' = 1_{a \times b}$$

2.2 Exercises for Good Preparation

2.2.1 Monic Arrows

Exercises from Goldblatt section 3.1: in any category, consider arrows $a \xrightarrow{f} b, b \xrightarrow{g} c$ as in the commutative diagram:

$$
\begin{tikzcd}
a \arrow[r, "f"] \arrow[dr, "g \circ f"'] & b \arrow[d, "g"] \\
& c
\end{tikzcd}
$$

(1) Prove that if f and g are both monic, then $g \circ f$ is monic.

> **Solution**
> Let $h, k : x \rightrightarrows a$ be such that $(g \circ f) \circ h = (g \circ f) \circ k$. By associativity and transitivity:
>
> $$g \circ (f \circ h) = (g \circ f) \circ h = (g \circ f) \circ k = g \circ (f \circ k)$$
>
> Since g is monic, it follows that:
> $$f \circ h = f \circ k$$
> Since f is monic, it follows that:
> $$h = k$$

(2) Prove that if $g \circ f$ is monic, then f is monic.

> **Solution**
> Let $h, k : x \rightrightarrows a$ be such that $f \circ h = f \circ k$. By associativity and substitution:
>
> $$(g \circ f) \circ h = g \circ (f \circ h) = g \circ (f \circ k) = (g \circ f) \circ k$$
>
> Since $g \circ f$ is monic, it follows that:
> $$h = k$$

2.2.2 Terminal Objects

Remaining exercise from Goldblatt section 3.6:

3. Show that an arrow $1 \to a$ whose domain is a terminal object must be monic.

> **Solution**
> Let $f : 1 \to a$ be an arrow. Let $h, k : x \rightrightarrows 1$ be such that $f \circ h = f \circ k$. Since h and k are already parallel arrows into a terminal object, it follows that $h = k$.
> Thus, f is monic.

2.2.3 Isomorphic Objects

Exercise 1 from Goldblatt section 3.4: For any $\mathscr{C}$-objects a, b, c in any category $\mathscr{C}$, prove the following:

(i) $a \cong a$

> **Solution**
> From Goldblatt section 3.3 exercise 1, $1_a : a \to a$ is an iso, and so witnesses that $a \cong a$.

(ii) if $a \cong b$, then $b \cong a$

> **Solution**
> If $a \cong b$ is witnessed by an iso $f : a \to b$, then by Goldblatt section 3.3 exercise 2, $f^{-1} : b \to a$ is an iso that witnesses $b \cong a$.

(iii) if $a \cong b$ and $b \cong c$, then $a \cong c$

> **Solution**
> If $a \cong b$ is witnessed by an iso $g : a \to b$ and if $b \cong c$ is witnessed by an iso $f : b \to c$, then by Goldblatt section 3.3 exercise 3, $f \circ g : a \to c$ is an iso that witnesses $a \cong c$.

2.2.4 Products

More exercises from Goldblatt section 3.8:

Exercise 7: Show that $(a \times b) \times c \cong a \times (b \times c)$.

Solution

It suffices to show that the following two arrows are inverses of each other:

$$\langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle\rangle : (a \times b) \times c \to a \times (b \times c)$$

$$\langle\langle \mathrm{pr}_a, \mathrm{pr}_b \circ \mathrm{pr}_{b\times c}\rangle, \mathrm{pr}_c \circ \mathrm{pr}_{b\times c}\rangle : a \times (b \times c) \to (a \times b) \times c$$

By computation, we have the following equalities:

$$\mathrm{pr}_a \circ \langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle\rangle = \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}$$

$$\mathrm{pr}_b \circ \mathrm{pr}_{b\times c} \circ \langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle\rangle = \mathrm{pr}_b \circ \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle = \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}$$

$$\mathrm{pr}_c \circ \mathrm{pr}_{b\times c} \circ \langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle\rangle = \mathrm{pr}_c \circ \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c \rangle = \mathrm{pr}_c$$

> By repeated use of Goldblatt exercises 1 and 3 of section 3.8, it follows that:
>
> $$\begin{aligned} &\langle\langle \mathrm{pr}_a, \mathrm{pr}_b \circ \mathrm{pr}_{b\times c}\rangle, \mathrm{pr}_c \circ \mathrm{pr}_{b\times c}\rangle \circ \langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c\rangle\rangle \\ &= \langle\langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}\rangle, \mathrm{pr}_c\rangle \\ &= \langle\langle \mathrm{pr}_a, \mathrm{pr}_b\rangle \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c\rangle \\ &= \langle 1_{a\times b} \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c\rangle \\ &= \langle \mathrm{pr}_{a\times b}, \mathrm{pr}_c\rangle \\ &= 1_{(a\times b)\times c} \end{aligned}$$
>
> A symmetric argument (with several computations omitted) would show that:
>
> $$\langle \mathrm{pr}_a \circ \mathrm{pr}_{a\times b}, \langle \mathrm{pr}_b \circ \mathrm{pr}_{a\times b}, \mathrm{pr}_c\rangle\rangle \circ \langle\langle \mathrm{pr}_a, \mathrm{pr}_b \circ \mathrm{pr}_{b\times c}\rangle, \mathrm{pr}_c \circ \mathrm{pr}_{b\times c}\rangle = 1_{a\times(b\times c)}$$

Exercise 8(i): Show that $(f \times h) \circ \langle g, k\rangle = \langle f \circ g, h \circ k\rangle$.

> **Solution**
>
> We may utilize previous exercises to compute as follows, (beginning with the expansion of $f \times g$):
>
> $$\begin{aligned} &(f \times h) \circ \langle g, k\rangle \\ &= \langle f \circ \mathrm{pr}_a, h \circ \mathrm{pr}_c\rangle \circ \langle g, k\rangle \\ &= \langle f \circ \mathrm{pr}_a \circ \langle g, k\rangle, h \circ \mathrm{pr}_c \circ \langle g, k\rangle\rangle \\ &= \langle f \circ g, h \circ k\rangle \end{aligned}$$

Exercise 8(ii): Show that $(f \times h) \circ (g \times k) = (f \circ g) \times (h \circ k)$.

> **Solution**
>
> We may utilize previous exercises to compute as follows:
>
> $$\begin{aligned} &(f \times h) \circ (g \times k) \\ &= \langle f \circ \mathrm{pr}_a, h \circ \mathrm{pr}_c\rangle \circ \langle g \circ \mathrm{pr}_e, k \circ \mathrm{pr}_{e'}\rangle \\ &= \langle f \circ \mathrm{pr}_a \circ \langle g \circ \mathrm{pr}_e, k \circ \mathrm{pr}_{e'}\rangle, h \circ \mathrm{pr}_c \circ \langle g \circ \mathrm{pr}_e, k \circ \mathrm{pr}_{e'}\rangle\rangle \\ &= \langle f \circ g \circ \mathrm{pr}_e, h \circ k \circ \mathrm{pr}_{e'}\rangle \\ &= (f \circ g) \times (h \circ k) \end{aligned}$$

2.3 Extra Exercises

2.3.1 Isomorphic Objects

Exercise 2 from Goldblatt section 3.4: Prove that $\mathbf{Finord}$ is a skeletal category.

($\mathbf{Finord}$ is the full subcategory of $\mathbf{Set}$ with finite ordinals as objects; refer to Goldblatt section 2.5 example 9 for further details if necessary.)

Solution

Suppose that $a \cong b$ are isomorphic objects of $\mathbf{Finord}$. As sets, they have the same cardinality $|a| = n = |b|$ for some finite n. The only object of $\mathbf{Finord}$ that has exactly n elements is $\{0, 1, \ldots, n-1\}$. Therefore, $a = \{0, 1, \ldots, n-1\} = b$.

Notes

It's worth noting that even though $a = b$, the isomorphism $a \cong b$ need not be 1_a; any permutation will do.

2.3.2 Products

Remaining exercises from Goldblatt section 3.8:

Exercise 9: Analyze in detail the formation of the projection arrows $\mathrm{pr}_1^m, \ldots, \mathrm{pr}_m^m$, and verify assertions relating to the last diagram (refer to Goldblatt). Show that for any product arrow

$$c \xrightarrow{\langle f_1, \ldots, f_m \rangle} a^m,$$

we have $\mathrm{pr}_j^m \circ \langle f_1, \ldots, f_m \rangle = f_j$, for $1 \leq j \leq m$.

Solution

For the sake of clarity, let us define inductively that $a^1 = a$ and that for $n \geq 1$, $a^{n+1} = a^n \times a$. For example, our convention means that $a^3 = (a \times a) \times a$.

Notes

The choice $a^{n+1} = a \times a^n$ for $n > 0$ would work as an alternative by symmetry, provided that we consistently adjust all details dependent on this alternative by the same symmetry.

We can define inductively that $\mathrm{pr}_1^1 = 1_a : \underbrace{a^1}_{a} \to a$ and that for $n \geq 1$ and $1 \leq j \leq n$,

$$\mathrm{pr}_{n+1}^{n+1} = \mathrm{pr}_a : \underbrace{a^{n+1}}_{a^n \times a} \to a$$

$$\mathrm{pr}_j^{n+1} = \mathrm{pr}_j^n \circ \mathrm{pr}_{a^n} : \underbrace{a^{n+1}}_{a^n \times a} \to a$$

Given arrows $f_1, \ldots, f_n : c \to a$, we may also define inductively that $\langle f_1 \rangle = f_1 : c \to \underbrace{a^1}_{a}$ and that for $n \geq 1$,

$$\langle f_1, \ldots, f_n, f_{n+1} \rangle = \langle \langle f_1, \ldots, f_n \rangle, f_{n+1} \rangle : c \to \underbrace{a^{n+1}}_{a^n \times a}$$

Showing the universal property of our proposed product diagram $(\mathrm{pr}_j^m)_{1 \leq j \leq n}$ is now an exercise in induction, for which we'll omit the details. Message us on Discord if you'd like the full version!

Showing that $\mathrm{pr}_j^m \circ \langle f_1, \ldots, f_m \rangle = f_j$ is also an exercise on induction (on m), which we'll go through now.

Base case $m = 1$:

We only have $j = 1$ to check. By definition,

$$\mathrm{pr}_1^1 \circ \langle f_1 \rangle = 1_a \circ f_1 = f_1$$

Inductive step $m = n + 1$ for $n \geq 1$:

If $j = n + 1$, then

$$\mathrm{pr}_{n+1}^{n+1} \circ \langle f_1, \ldots, f_n, f_{n+1} \rangle = \mathrm{pr}_a \circ \langle \langle f_1, \ldots, f_n \rangle, f_{n+1} \rangle = f_{n+1}$$

If $j \leq n$,

$$\mathrm{pr}_j^{n+1} \circ \langle f_1, \ldots, f_n, f_{n+1} \rangle = \mathrm{pr}_j^n \circ \mathrm{pr}_{a^n} \circ \langle \langle f_1, \ldots, f_n \rangle, f_{n+1} \rangle = f_{n+1} = \mathrm{pr}_j^n \circ \langle f_1, \ldots, f_n \rangle = f_j$$

where the last step uses the inductive hypothesis.

Exercise 10: Develop the notion of the product $a_1 \times a_2 \times \cdots \times a_m$ of m objects (possibly different) and the product of m arrows (possibly different).

Solution

Refer to Meeting III slides for a construction of the product of n (or more objects), possibly all different. If further details are desired, message us on Discord!

2.3.3 Co-products

Exercises from Goldblatt section 3.9:

Exercise 1: Show that $A + B, i_A, i_B$ as just defined (in the category $\mathbf{Set}$, refer to Goldblatt) satisfy the co-product definition.

> **Solution**
>
> Existance of universal property:
> Let $f : A \to C$ and $g : B \to C$ be two arbitrary functions. Let us define $[f, g] : A + B \to C$ as follows:
>
> $$[f,g](x) = \begin{cases} f(\mathrm{pr}_A(x)) & \text{if } x \in A \times \{0\} \\ g(\mathrm{pr}_B(x)) & \text{if } x \in B \times \{1\} \end{cases}$$
>
> Note that $A \times \{0\}$ is disjoint from $B \times \{1\}$ since $\langle a, 0\rangle \neq \langle b, 1\rangle$ for any combination of $a \in A$ and $b \in B$. Thus, our proposed definition is single-valued.
> By computation:
>
> $$\begin{aligned} ([f,g] \circ i_A)(a) &= [f,g](\langle a, 0\rangle) = f(\mathrm{pr}_A(\langle a, 0\rangle)) = f(a) && \forall a \in A \\ ([f,g] \circ i_B)(b) &= [f,g](\langle b, 1\rangle) = g(\mathrm{pr}_B(\langle b, 1\rangle)) = g(b) && \forall b \in B \end{aligned}$$
>
> This shows the existence aspect of the universal property of a co-product.
> Uniqueness of universal property:
> Let $h, k : A + B \rightrightarrows C$ be such that $h \circ i_A = k \circ i_B$ and $h \circ i_B = k \circ i_B$. Let $x \in A + B$. There are two cases to consider:
>
> - If $x \in A \times \{0\}$, then $x = i_A(a)$ for some $a \in A$ because the image of i_A is precisely $A \times \{0\}$. Thus, $h(x) = h(i_A(a)) = k(i_A(a)) = k(x)$
> - If $x \in B \times \{1\}$, then $x = i_B(b)$ for some $b \in B$ because the image of i_B is precisely $B \times \{1\}$. Thus, $h(x) = h(i_B(b)) = k(i_B(b)) = k(x)$
>
> Thus, we can conclude that $h = k$.

Exercise 2: If $A \cap B = \emptyset$, show $A \cup B \cong A + B$.

> **Solution**
>
> Let $f : A \hookrightarrow A \cup B$ and $g : B \hookrightarrow A \cup B$ be the two (monic) inclusion arrows. We claim that the morphism $[f, g] : A + B \to A \cup B$ is an isomorphism. It suffices to check that $[f, g]$ is injective and surjective.
> Surjectivity:
> Let $x \in A \cup B$. If $x \in A$, then $[f,g](\langle x, 0\rangle) = f(x) = x$. If $x \in B$, then $[f,g](\langle x, 1\rangle) = g(x) = x$. Thus, f is surjective.
> Injectivity:
> Suppose $\langle x, m\rangle, \langle y, n\rangle \in A + B$ are such that $[f,g](\langle x, m\rangle) = [f,g](\langle y, n\rangle)$. We have four cases

> to analyze:
>
> - If $m = 0$ and $n = 1$, then $f(x) = [f, g](\langle x, m\rangle) = [f, g](\langle y, n\rangle) = g(y)$. But since $f(x) \in A$ and $g(y) \in B$, this contradicts $A \cap B = \emptyset$.
> - Similarly, if $m = 1$ and $n = 0$, then $g(x) = [f, g](\langle x, m\rangle) = [f, g](\langle y, n\rangle) = f(y)$. But since $g(x) \in B$ and $f(y) \in A$, this contradicts $A \cap B = \emptyset$.
> - If $m = n = 0$, then $f(x) = [f, g](\langle x, m\rangle) = [f, g](\langle y, n\rangle) = f(y)$. Since f is injective, it follows that $x = y$.
> - If $m = n = 1$, then $g(x) = [f, g](\langle x, m\rangle) = [f, g](\langle y, n\rangle) = g(y)$. Since g is injective, it follows that $x = y$.
>
> In all cases, we can conclude $x = y$. Thus, $[f, g]$ is injective.

Exercise 3: Define the co-product arrow $f + g : a + b \to c + d$ of arrows $f : a \to c$ and $g : b \to d$ and dualise all of the Exercises in section 3.8 (see Goldblatt).

(Write out the dual statement of each exercise; you do not need to write out a dual proof. Though it can feel tedious, this exercise helps develop intuition about dual concepts. It can be insightful to see how a concept and its dual may be fundamentally different constructions, yet behave categorically similar.)

> **Solution**
>
> The arrow $f + g : a + b \to c + d$ can be analogously defined as:
>
> $$f + g = [i_c \circ f, i_d \circ g]$$
>
> Here is the analagous list of properties that various aspects of the co-product satisfy:
>
> - $[i_a, i_b] = 1_{a+b}$
> - If $[f, g] = [h, k]$, then $f = k$ and $g = h$
> - $[e \circ f, e \circ g] = e \circ [f, g]$
> - If $\mathscr{C}$ has an initial object and coproducts, then for any $\mathscr{C}$-object a, $a \cong a + 0$ and indeed $[1_a, 0_a]$ is iso.
> - $1_a + 1_b = 1_{a+b}$
> - If $a + b$ exists, then $b + a$ also exists and that $a + b \cong b + a$
> - $(a + b) + c \cong a + (b + c)$
> - $[f, g] \circ (h + k) = [f \circ h, g \circ k]$
> - $(f + h) \circ (g + k) = (f \circ h) + (h \circ k)$

2.3.4 Jointly Monic Pair

Let a, b be $\mathscr{C}$-objects. Suppose that we have a "span from a to b", which is a diagram in $\mathscr{C}$ of the following shape:

$$\begin{tikzcd} & r \arrow[dl, "f"'] \arrow[dr, "g"] & \\ a & & b \end{tikzcd}$$

We say that $\{f, g\}$ is **jointly monic** if for all parallel pairs $h, k : x \to r$, the following holds:

$$\textit{whenever } f \circ h = f \circ k \text{ and } g \circ h = g \circ k, \text{ then } h = k$$

This is a generalization of being "left-cancellable" in the sense that both conditions need to hold true before we can simultaneously cancel out f and g.

Exercise: Suppose that the product $a \times b$ exists. Prove that $\langle f, g \rangle : r \to a \times b$ is monic if and only if $\{f, g\}$ is jointly monic.

Solution

Forward direction:
Suppose that $\langle f, g \rangle : r \to a \times b$ is monic. Let $h, k : x \rightrightarrows r$ be arrows such that $f \circ h = f \circ k$ and $g \circ h = g \circ k$. Then, twice utilizing exercise 3 from Goldblatt section 3.8, we have that:

$$\langle f, g \rangle \circ h = \langle f \circ h, g \circ h \rangle = \langle f \circ k, g \circ k \rangle = \langle f, g \rangle \circ k$$

Since $\langle f, g \rangle$ is monic, it follows that $h = k$.
Reverse direction:
Suppose that $\{f, g\}$ is jointly monic. Let $h, k : x \rightrightarrows r$ be such that $\langle f, g \rangle \circ h = \langle f, g \rangle \circ k$. Utilizing exercise 3 from Goldblatt section 3.8, we have that:

$$\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h = \langle f, g \rangle \circ k = \langle f \circ k, g \circ k \rangle$$

Utilizing exercise 2 from Goldblatt section 3.8, we have $f \circ h = f \circ k$ and $g \circ h = g \circ k$.
Since $\{f, g\}$ is jointly monic, it follows that $h = k$.

Meeting IV

Category Theory Pt. 3

3.1 Minimal Exercises

3.1.1 Equalizers

Exercises from Goldblatt section 3.10:

Let $E \overset{f}{\rightarrowtail} A$ be a monomorphism (a.k.a. monic arrow) in **Set**. Define $h, g : A \to \{0, 1\}$ as follows:

$$h(x) = 1$$
$$g(x) = \begin{cases} 1 & \text{if } x \in \operatorname{Im} f \\ 0 & \text{if } x \notin \operatorname{Im} f \end{cases}$$

Exercise 1: Prove that f is the equalizer of g and h.

> **Solution**
>
> Existence:
>
> > **Solution**
> >
> > Let $t : X \to A$ be an arbitrary function such that $g \circ t = h \circ t$. Thus, $g(t(x)) = 1$ for all $x \in X$. By definition, this means $t(x) \in \operatorname{Im} f$. Since f is a monomorphism (and hence injective in $\mathbf{Set}$), this implies there is exactly one such $e_x \in E$ with $f(e_x) = t(x)$. Let $\hat{t} : X \to E$ be defined as $t(x) = e_x$. Then, $f(\hat{t}(x)) = f(e_x) = t(x)$
>
> Uniqueness:
>
> > **Solution**
> >
> > Since f is a monomorphism, the uniqueness aspect of the universal property follows. For, if $\hat{t}' : X \to A$ were another function such that $f \circ \hat{t} = f \circ \hat{t}'$, then $\hat{t} = \hat{t}'$ by the left-cancellability of f.

Exercise 2: Show that in a poset, the only equalizers are the identity arrows.

Solution

In a poset, every morphism is both monic and epic. (This is because parallel maps in a poset coincide regardless of assumption.)
Therefore, any equalizer would be epic. As epic equalizers are isos (by theorem 2 of Goldblatt section 3.10) and isos are identities in a poset, it follows that equalizers in a poset are identities.

3.1.2 Pullbacks

Exercise from Goldblatt section 3.13: Show that if

$$\begin{tikzcd} a \arrow[r, "g"] \arrow[d] & b \arrow[d] \\ c \arrow[r, "f"] & d \end{tikzcd}$$

is a pullback square, and f is monic, then g is also monic.

Solution

For ease of reference, let the unnamed arrows be $h : b \to d$ and $k : a \to c$ as in

$$\begin{tikzcd} a \arrow[r, "g"] \arrow[d, "k"'] & b \arrow[d, "h"] \\ c \arrow[r, "f"] & d \end{tikzcd}$$

Let $p, q : x \rightrightarrows a$ be such that $g \circ p = g \circ q$.
Firstly, notice $f \circ k \circ p = h \circ g \circ p = h \circ g \circ q = f \circ k \circ q$. Therefore by monicity of f, it follows $k \circ p = k \circ q$.
Secondly, we can combine the just-proven fact that $k \circ p = k \circ q$ with the assumption that $g \circ p = g \circ q$ in order to utilize the uniqueness aspect of the universal property of pullback. Therefore, we can conclude that $p = q$.

3.1.3 Subobjects

Exercises from Goldblatt section 4.1:

Exercise 1: In $\mathbf{Set}$, verify that $\mathrm{Sub}(D) \cong \mathscr{P}(D)$.

Solution

For each subset $S \subseteq D$, associate with it the subobject corresponding to the evident inclusion function $(1_D)|_S : S \hookrightarrow D$. Let us show that this association is both injective and surjective.
Injectivity:

> > **Solution**
> >
> > Let $S, T \subseteq D$ two subsets such that $[1_D)|_S : S \hookrightarrow D] = [1_D)|_T : T \hookrightarrow D]$ as subobjects.
> > Let $f : S \xrightarrow{\cong} T$ be such that $(1_D)|_T \circ f = (1_D)|_S$, thereby witnessing that the two subobjects are identical.
> > For each $s \in S$, this implies that $f(s) = (1_D)|_T(f(s)) = (1_D)|_S(s) = s$. Therefore, $s = f(s) \in T$. Thus, $S \subseteq T$.
> > By symmetric reasoning, $T \subseteq S$. Therefore, $S = T$.
>
> Surjectivity:
>
> > **Solution**
> >
> > Let $[m : A \rightarrowtail D]$ be any subobject, say with representative m. Since m is injective, it induces an isomorphism onto its image $\operatorname{Im} m \subseteq D$. Call the isomorphism $\hat{m} : A \xrightarrow{\cong} \operatorname{Im} m$, which satisifies $\hat{m}(a) = m(a)$ by definition for each $a \in A$.
> > Observe that $(1_D)|_{\operatorname{Im} m}(\hat{m}(a)) = 1_D(\hat{m}(a)) = \hat{m}(a) = m(a)$ for each $a \in A$. Thus, we have a commutative triangle
> >
> > $$\begin{tikzcd} A \arrow[rr, "\hat{m}", "\cong"'] \arrow[dr, "m"'] & & \operatorname{Im} m \arrow[dl, "(1_D)|_{\operatorname{Im} m}"] \\ & D & \end{tikzcd}$$
> >
> > Therefore, $[m] = [(1_D)|_{\operatorname{Im} m}]$ as subobjects

Exercise 2: Prove that $ev \circ \langle \ulcorner f \urcorner, x \rangle = f \circ x$ (Refer to Goldblatt for notation and context.)

If you have not read section 3.16 on exponentials, you may do exercise 2 just for **Set**.

> **Solution**
>
> By calculation while freely using results from Goldblatt section 3.8, we have the following chain of equalities:
>
> $$\begin{aligned} & ev \circ \langle \ulcorner f \urcorner, x \rangle \\ &= ev \circ \langle \ulcorner f \urcorner \circ 1_1, 1_a \circ x \rangle \\ &= ev \circ (\ulcorner f \urcorner \times 1_a) \circ \langle 1_1, x \rangle \\ &= f \circ \mathrm{pr}_a \circ \langle 1_1, x \rangle \\ &= f \circ x \end{aligned}$$

3.1.4 Classifying subobjects

Exercises from Goldblatt section 4.2:

Exercise 1: Show that the character of $true : 1 \rightarrowtail \Omega$ is 1_Ω

$$\begin{tikzcd} 1 \arrow[r, tail, "true"] \arrow[d] & \Omega \arrow[d, "1_\Omega"] \\ 1 \arrow[r, "true"] & \Omega \end{tikzcd}$$

i.e. $\chi_{true} = 1_\Omega$.

Solution

The square commutes by the identity axioms of a category.
The left and right arrows are both monic with the left arrow being an iso. Thus, the conditions for 3.3.1 exercise 2 are readily verified. It follows that the square is a pullback.
The conclusion follows by definition of the Ω-Axiom.

Exercise 2: Show that $\chi_{1_\Omega} = true_\Omega = true \circ !_\Omega$.

$$\begin{tikzcd} \Omega \arrow[r, tail, "1_\Omega"] \arrow[d, "!_\Omega"'] & \Omega \arrow[d, "true \circ !_\Omega"] \\ 1 \arrow[r, "true"] & \Omega \end{tikzcd}$$

Solution

The square commutes by the identity axioms of a category.
The top and bottom arrows are both monic with the top arrow being an iso. Thus, the conditions for 3.3.1 exercise 2 are readily verified. It follows that the square is a pullback.
The conclusion follows by definition of the Ω-Axiom.

Exercise 3: Show that for any $f : a \to b$,

$$\begin{tikzcd} a \arrow[rr, "f"] \arrow[dr, "true_a"'] & & b \arrow[dl, "true_b"] \\ & \Omega & \end{tikzcd}$$

$true_b \circ f = true_a$.

Solution

If suffices to show that both sides classify the same subobject.

Notice that we have the following arrangement of pullback diagrams:

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[d, "1_a"] \arrow[dr, phantom, "\lrcorner", very near start] & b \arrow[r, "!"] \arrow[d, "1_b"] \arrow[dr, phantom, "\lrcorner", very near start] & 1 \arrow[r, "1_1"] \arrow[d, "1_1"] \arrow[dr, phantom, "\lrcorner", very near start] & 1 \arrow[d, "true"] \\ a \arrow[r, "f"] & b \arrow[r, "!"] & 1 \arrow[r, "true"] & \Omega \end{tikzcd}$$

The first and second are pullbacks by application of 3.3.1 exercise 2. The last is a pullback because $true$ is monic.
We also have the following arrangement of pullback diagrams:

$$\begin{tikzcd} a \arrow[r, "!"] \arrow[d, "1_a"] \arrow[dr, phantom, "\lrcorner", very near start] & 1 \arrow[r, "1_1"] \arrow[d, "1_1"] \arrow[dr, phantom, "\lrcorner", very near start] & 1 \arrow[d, "true"] \\ b \arrow[r, "!"] & 1 \arrow[r, "true"] & \Omega \end{tikzcd}$$

The first is a pullback by application of 3.3.1 exercise 2. The last is a pullback because $true$ is monic.
By repreated application of the two pullback lemma, the outer rectangles are pullback diagrams in each case. Since we have expressed the subobject for 1_a as a pullback of $true$ along two paths, it follows both paths have equal composites.

3.2 Exercises for Good Preparation

3.2.1 Pushouts

Exercise from Goldblatt section 3.14: Dualise section 3.13.

(In particular, write out the concept of a pushout and write out the duals of examples 5 through 9, especially example 8 regarding the dual of the pullback lemma.)

Solution

Ex 5. In a pre-order $(P, \sqsubseteq)$,

$$\begin{tikzcd} s \arrow[r] \arrow[d] & q \arrow[d] \\ p \arrow[r] & r \end{tikzcd}$$

is a pushout square iff r is a co-product of p and q.

> **Notes**
> A co-product in a pre-order is the same as a least upper bound.

Ex 6. In any category with an initial object, if

$$\begin{tikzcd} 0 \arrow[r, "0_b"] \arrow[d, "0_a"'] & b \arrow[d, "g"] \\ a \arrow[r, "f"] & c \end{tikzcd}$$

is a pushout, then (f, g) is a co-product (l.u.b.) of a and b.

Ex 7. In any category, if

$$\begin{tikzcd} d \arrow[r, "g"] \arrow[d, "f"'] & a \arrow[d, "q"] \\ a \arrow[r, "q"] & c \end{tikzcd}$$

is a pushout, then q is a co-equalizer of f and g.

Ex 8. The Pushout Lemma. If a diagram of the form

$$\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \arrow[r] & \bullet \end{tikzcd}$$

commutes, then

(i) if the two small squares are pushouts, then the outer “rectangle” (with top and bottom edges the evident composites) is a pushout;

(ii) if the outer rectangle and the left hand square are pushouts, then so is the right hand square.

Ex 9. In any category, an arrow $f : a \to b$ is epic iff

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[d, "f"'] & b \arrow[d, "1_b"] \\ b \arrow[r, "1_b"] & b \end{tikzcd}$$

is a pushout square.

3.2.2 Completeness

Exercises from Goldblatt section 3.15:

(1) Verify (B), and consider the details of that construction in **Set**. From Goldblatt, (B) states: given pullbacks and products, from a parallel pair $f, g : a \rightrightarrows b$ and a pullback of the form

$$\begin{tikzcd} d \arrow[r, "q"] \arrow[d, "p"'] & a \arrow[d, "\langle 1_a, g\rangle"] \\ a \arrow[r, "\langle 1_a, f\rangle"'] & a \times b, \end{tikzcd}$$

it follows readily (from section 3.8) that $p = q$, and that this arrow is an equaliser of f and g.

> **Solution**
> Suppose the above square is a pullback. By calculation, we have that:
>
> $$p = 1_a \circ p = \mathrm{pr}_a \circ \langle 1_a, f\rangle \circ p = \mathrm{pr}_a \circ \langle 1_a, g\rangle \circ q = 1_a \circ q = q$$
>
> Next, suppose that $h : x \to a$ is any arrow such that $f \circ h = g \circ h$. Then,
>
> $$h \circ \langle 1_a, f\rangle = \langle h \circ 1_a, h \circ f\rangle = \langle h \circ 1_a, h \circ g\rangle = h \circ \langle 1_a, g\rangle$$
>
> Hence, by definition of pullback, there is a unique $u : x \to d$ such that $p \circ u = h$ and $q \circ u = h$.
> Equivalently since $p = q$, there is a unique $u : x \to d$ such that $p \circ u = h$. This verifies the universal property for p to be an equalizer.
> In $\mathbf{Set}$, the construction computes
>
> $$\begin{aligned} d &= \{\langle x, y\rangle \in a \times a : \langle x, f(x)\rangle = \langle y, f(y)\rangle\} \\ &= \{\langle x, y\rangle \in a \times a : x = y \wedge f(x) = g(y)\} \\ &= \{\langle x, x\rangle \in a \times a : f(x) = g(x)\} \\ &\cong \{x \in a : f(x) = g(x)\} \end{aligned}$$

(2) Show how to construct pullbacks from products and equalisers. (Refer to Goldblatt for a hint.)

> **Solution**
> Let
>
> $$\begin{tikzcd} & b \arrow[d, "g"] \\ a \arrow[r, "f"] & c \end{tikzcd}$$
>
> be given.

Consider an equalizer diagram of the following form:

$$\begin{tikzcd} e \arrow[r, "s"] & a \times b \arrow[r, shift left, "f \circ \mathrm{pr}_a"] \arrow[r, shift right, "g \circ \mathrm{pr}_b"'] & c \end{tikzcd}$$

Let $p = \mathrm{pr}_a \circ s$ and $q = \mathrm{pr}_b \circ s$. Then we claim that p and q fit in a pullback square as follows:

$$\begin{tikzcd} e \arrow[r, "q"] \arrow[d, "p"] \arrow[dr, phantom, "\lrcorner", very near start] & b \arrow[d, "g"] \\ a \arrow[r, "f"] & c \end{tikzcd}$$

Let us verify the three required conditions of being a pullback diagram.
Commutativity of the diagram:

> **Solution**
>
> By calculation, we have the following:
>
> $$f \circ p = f \circ \mathrm{pr}_a \circ s = g \circ \mathrm{pr}_b \circ s = g \circ q$$

Uniqueness aspect of the universal property:

> **Solution**
>
> Suppose $u, v : x \rightrightarrows e$ are such that $p \circ u = p \circ v$ and $q \circ u = q \circ v$. Then by calculation:
>
> $$\begin{aligned} & s \circ u \\ &= 1_{a \times b} \circ s \circ u \\ &= \langle \mathrm{pr}_a, \mathrm{pr}_b \rangle \circ s \circ u \\ &= \langle \mathrm{pr}_a \circ s, \mathrm{pr}_b \circ s \rangle \circ u \\ &= \langle p, q \rangle \circ u \\ &= \langle p \circ u, q \circ u \rangle \\ &= \langle p \circ v, q \circ v \rangle \\ &= \langle p, q \rangle \circ v \\ &= \langle \mathrm{pr}_a \circ s, \mathrm{pr}_b \circ s \rangle \circ v \\ &= \langle \mathrm{pr}_a, \mathrm{pr}_b \rangle \circ s \circ v \\ &= 1_{a \times b} \circ s \circ v \\ &= s \circ v \end{aligned}$$
>
> Since s is an equalizer, it is a monomorphism. Therefore, $u = v$.

Existence aspect of the universal property:

Solution

Let $h : x \to a$ and $k : x \to b$ be such that $f \circ h = g \circ k$. Then by calculation:

$$\begin{aligned} & f \circ \mathrm{pr}_a \circ \langle h, k \rangle \\ = \; & f \circ h \\ = \; & g \circ k \\ = \; & g \circ \mathrm{pr}_b \circ \langle h, k \rangle \end{aligned}$$

By definition of s being an equalizer of $f \circ \mathrm{pr}_a$ and $g \circ \mathrm{pr}_b$, it follows that there exists a morphism $u : x \to e$ such that $\langle h, k \rangle = s \circ u$.
We calculate the following:

$$p \circ u = \mathrm{pr}_a \circ s \circ u = \mathrm{pr}_a \circ s \langle h, k \rangle = h$$

and:

$$q \circ u = \mathrm{pr}_b \circ s \circ u = \mathrm{pr}_b \circ s \langle h, k \rangle = k$$

Thus, u is a valid cone morphism.

(3) Dualise the Theorem of this section. (Refer to Goldblatt. You only need to write out the dual statement of the theorem.)

Solution

Theorem: If $\mathscr{C}$ has an initial object, and a pushout for each pair of $\mathscr{C}$-arrows with common domain, then $\mathscr{C}$ is finitely co-complete.
(A) Given an initial object and pushouts, the co-product of a and b is got from the pushout of $a \leftarrow 0 \to b$;
(B) given pushouts and co-products, from a parallel pair $f, g : a \rightrightarrows b$ we first form the co-product arrows

$$a + b \xrightarrow{[f, 1_b]} b \quad \text{and} \quad a + b \xrightarrow{[g, 1_b]} b$$

and then their pushout

$$\begin{tikzcd} a + b \arrow[r, "{[f, 1_b]}"] \arrow[d, "{[g, 1_b]}"'] & b \arrow[d, "i"] \\ b \arrow[r, "j"] & q \end{tikzcd}$$

It follows readily that $i = j$, and that this arrow is a co-equalizer of f and g.

3.2.3 Exponentiation

Exercise from Goldblatt section 3.16: Prove part (5) of the Theorem, and interpret (1)–(5) as they apply to **Set**. For reference, the theorem is:

> **Theorem**
> Let $\mathscr{C}$ be a Cartesian closed category with an initial object 0. Then in $\mathscr{C}$,
> (1) $0 \cong 0 \times a$, for any object a;
> (2) if there exists an arrow $a \to 0$, then $a \cong 0$;
> (3) if $0 \cong 1$, then the category $\mathscr{C}$ is degenerate, i.e. all $\mathscr{C}$-objects are isomorphic;
> (4) any arrow $0 \to a$ with dom 0 is monic;
> (5) $a^1 \cong a, a^0 \cong 1, 1^a \cong 1$.

(1) $0 \cong 0 \times a$, for any object a;

> **Solution**
> In $\mathbf{Set}$, a Cartesian product involving the empty set is the empty set.

(2) if there exists an arrow $a \to 0$, then $a \cong 0$;

> **Solution**
> In $\mathbf{Set}$, the only arrows into the empty set are those that have empty domain.

(3) if $0 \cong 1$, then the category $\mathscr{C}$ is degenerate, i.e. all $\mathscr{C}$-objects are isomorphic;

> **Solution**
> $\mathbf{Set}$ is not degenerate. The empty set is not isomorphic to a singleton set.

(4) any arrow $0 \to a$ with dom 0 is monic;

> **Solution**
> In $\mathbf{Set}$, every empty function is monic, since all parallel arrows into the empty domain coincide.

(5) $a^1 \cong a, a^0 \cong 1, 1^a \cong 1$.

Solution

In $\mathbf{Set}$, there are just as many functions $1 \to a$ as there are elements of a. There's exactly one function from the empty set to a. There's exactly one function from a to a singleton set.

Solution — $a^1 \cong a$

We claim that $\mathrm{pr}_a : a \times 1 \to a$ satisfies the universal property of exponentiation.
Uniqueness:
This follows because pr_a is an isomorphism.
Existence:
Let $f : x \times 1 \to a$ be any map. Then by calculation,

$$\begin{aligned} &\mathrm{pr}_a \circ ((f \circ \langle 1_x, !_x \rangle) \times 1_1) \\ &= (f \circ \langle 1_x, !_x \rangle) \circ \mathrm{pr}_x \\ &= f \circ \langle 1_x \circ \mathrm{pr}_x, !_x \circ \mathrm{pr}_x \rangle \\ &= f \circ \langle \mathrm{pr}_x, !_1 \rangle \\ &= f \circ \langle \mathrm{pr}_x, \mathrm{pr}_1 \rangle \\ &= f \circ 1_{x \times 1} \\ &= f \end{aligned}$$

Solution — $a^0 \cong 1$

We claim that $0_a \circ \mathrm{pr}_0 : 1 \times 0 \to a$ satisfies the universal property of exponentiation.
Uniqueness:
This follows because maps from a fixed domain into 1 are unique.
Existence:
Let $f : x \times 0 \to a$ be any map. By the theorem part (2), since $x \times 0$ admits a projection map to 0, it follows that $x \times 0$ is initial.
Consider the map $!_x : x \to 1$. Note that $0_a \circ \mathrm{pr}_0 \circ (!_x \times 1_0) : x \times 0 \to a$ is parallel with $f : x \times 0 \to a$.
Since $x \times 0$ is initial, parallel outgoing maps coincide. Therefore,

$$0_a \circ \mathrm{pr}_0 \circ (!_x \times 1_0) = f$$

> **Solution —** $1^a \cong 1$
>
> We claim that $!_a : 1 \times a \to 1$ satisfies the universal property of exponentiation.
> Uniqueness:
> This follows because maps from a fixed domain into 1 are unique.
> Existence:
> Let $f : x \times a \to 1$ be any map. Consider the map $!_x : x \to 1$. Note that $!_a \circ (!_x \times 1_a) : x \times a \to 1$ is parallel with $f : x \times a \to 1$.
> Since 1 is terminal, parallel incoming maps coincide. Therefore,
>
> $$!_a \circ (!_x \times 1_a) = f$$

3.2.4 Topos

Exercise 1: Prove that in a topos:

- $\mathrm{Sub}(0) \cong 1$

> **Solution**
>
> From Goldblatt section 4.2, we have the isomorphism $\mathrm{Sub}(0) \cong \mathscr{C}(0, \Omega)$. Since 0 is an initial object, there is exactly one arrow $0 \to \Omega$.
> In other words, $\mathscr{C}(0, \Omega) \cong 1$. The result follows by transitivity of being isomorphic.

- $\mathrm{Sub}(A + B) \cong \mathrm{Sub}(A) \times \mathrm{Sub}(B)$

> **Solution**
>
> From Goldblatt section 4.2, we have the isomorphism $\mathrm{Sub}(A+B) \cong \mathscr{C}(A+B, \Omega)$. Since $A+B$ is a co-product, there is exactly one arrow $A+B \to \Omega$ for each pair of arrows $A \to \Omega$ and $B \to \Omega$. Thus, $\mathscr{C}(A+B, \Omega) \cong \mathscr{C}(A, \Omega) \times \mathscr{C}(B, \Omega)$.
> Again from Goldblatt section 4.2, we have the isomorphisms $\mathscr{C}(A, \Omega) \cong \mathrm{Sub}(A)$ and $\mathscr{C}(B, \Omega) \cong \mathrm{Sub}(B)$. The result follows by transitivity of being isomorphic.

If you need a hint for the second one, message us on Discord!

Exercise 2: in the topos $\mathbf{Set}^{\to}$, find the characteristic arrow of the following monic (from the left arrow to the right arrow):

$$\begin{tikzcd} \{0\} \arrow[r, hook] \arrow[d, hook] & \{0,1,2\} \arrow[d, hook] \\ \{0,1\} \arrow[r, hook] & \{0,1,2,3\} \end{tikzcd}$$

Solution

Let $\psi : \{0, 1, 2\} \to \{0, \frac{1}{2}, 1\}$ be the map such that $\psi(0) = 1$ and $\psi(1) = \frac{1}{2}$ and $\psi(2) = 0$.
Let $\chi : \{0, 1, 2, 3\} \to \{0, 1\}$ be the map such that $0 \mapsto 1$ and $1 \mapsto 1$ and $2 \mapsto 0$ and $3 \mapsto 0$.
Then, we have the following morphism in $\mathbf{Set}^{\to}$ from the left arrow to the right arrow:

$$\begin{tikzcd}
\{0, 1, 2\} \arrow[r, "\psi"] \arrow[d, hook] & \{0, \frac{1}{2}, 1\} \arrow[d, "t"] \\
\{0, 1, 2, 3\} \arrow[r, "\chi"] & \{0, 1\}
\end{tikzcd}$$

This morphism is the characteristic arrow according to the procedure specified in example 5 of Goldblatt section 4.4.

3.3 Extra Exercises

3.3.1 Pullbacks

Suppose

$$\begin{tikzcd}
d \arrow[r, "q"] \arrow[d, "p"'] & b \arrow[d, "g"] \\
a \arrow[r, "f"'] & c
\end{tikzcd}$$

is a commutative diagram.

Exercise 1: Suppose the above is a pullback square. Prove that $\{p, q\}$ is jointly monic (refer to exercise 2.3.4 above for the definition).

> **Solution**
>
> Let $u, v : x \rightrightarrows d$ be such that $p \circ u = p \circ v$ and $q \circ u = q \circ v$.
> Let $s : x \to a$ be the common composite $s = p \circ u = p \circ v$. Let $t : x \to b$ be the common composite $q \circ u = q \circ v$.
> Note $g \circ s = g \circ p \circ u = f \circ p \circ u = f \circ t$. Therefore, s and t form a commutative square with f and g.
> By definition of pullback, there is a unique cone morphism from $\langle f, g \rangle$ to $\langle p, q \rangle$. Both u and v satisfy this criteria. Therefore, $u = v$.

Exercise 2: Suppose that p and g are both monic. Prove that the above square is a pullback iff for all commutative diagrams of the form

$$\begin{tikzcd}
x \arrow[rrd, bend left, "t"] \arrow[rdd, bend right, "s"'] & & \\
& & b \arrow[d, "g"] \\
& a \arrow[r, "f"] & c
\end{tikzcd},$$

there exists an arrow $x \to d$ making the following diagram commute:

$$\begin{tikzcd}
x \arrow[rd] \arrow[rdd, bend right, "s"'] & \\
& d \arrow[d, "p"'] \\
& a
\end{tikzcd}$$

> **Solution**
>
> We are given that the diagram commutes, hence it corresponds to a cone. It suffices to prove that the cone satisfies the universal property of being a limiting cone.
> Suppose we have another cone: i.e. let $s : x \to a$ and $t : x \to b$ be such that $f \circ s = g \circ t$.
> Uniqueness:
>
> > **Solution**
> >
> > Let $u, v : x \rightrightarrows d$ be such that $p \circ u = s = p \circ v$ and $q \circ u = t = q \circ v$. Since we are given that p is monic, we utilize that $p \circ u = p \circ v$ to prove that $u = v$.
>
> Existence:
>
> > **Solution**
> >
> > Let $u : x \to d$ be any arrow such that $p \circ u = s$; this exists by assumption.
> > Firstly, notice that $g \circ q \circ u = f \circ p \circ u = f \circ s = g \circ t$. Since we are given that g is monic, it follows that $q \circ u = g$.
> > Combined with the fact that $p \circ u = s$ by assumption, it follows that $u : x \to d$ is a morphism of cones.

> **Theorem — Corollary**
>
> Suppose that g is monic and p is iso. The above square is then a pullback.

> **Solution — Corollary**
>
> Let $s : x \to a$ and $t : x \to b$ be such that $f \circ s = g \circ t$.
> Since $p \circ p^{-1} \circ s = 1_a \circ s = s$, it follows that $p^{-1} \circ s$ witnesses the hypothesis of the above exercise. The conclusion follows.

Exercise 3: Let $\mathscr{C}$ be a category with pullbacks. Prove that $\mathscr{C}$ has limits of the following shape:

$$\begin{tikzcd}
\bullet \arrow[dr] & & \bullet \arrow[dl] \arrow[dr] & & \bullet \arrow[dl] \\
& \bullet & & \bullet &
\end{tikzcd}$$

> **Proof sketch**

> Suppose we have the following diagram:
>
> $$\begin{tikzcd} A \arrow[dr, "f"'] & & S \arrow[dl, "c"] \arrow[dr, "d"'] & & B \arrow[dl, "g"] \\ & C & & D & \end{tikzcd}$$
>
> Let us form three pullbacks as follows:
>
> $$\begin{tikzcd} & & R \arrow[dl, "p"'] \arrow[dr, "q"] \arrow[dd, phantom, "\vee", very near start] & & \\ & P \arrow[dl, "r"'] \arrow[dr, "s"] \arrow[dd, phantom, "\vee", very near start] & & Q \arrow[dl, "t"'] \arrow[dr, "u"] \arrow[dd, phantom, "\vee", very near start] & \\ A \arrow[dr, "f"'] & & S \arrow[dl, "c"] \arrow[dr, "d"'] & & B \arrow[dl, "g"] \\ & C & & D & \end{tikzcd}$$
>
> We claim that R is the tip of our cone with the evident cone projections. To verify that the evident cone projections satisfy the constraints of our starting diagram, it is sufficient to observe that the augmented diagram is commutative.
> In particular, let $a = r \circ p : R \to A$ and $b = u \circ q$ and $e = s \circ p = t \circ q : R \to S$. These morphisms determine the rest of the cone legs since $f \circ a = c \circ e : R \to C$ and $d \circ e = g \circ b : R \to D$.
> It is an exercise in diagram chasing to verify that this proposed cone is indeed a limiting cone.
> If you'd like the remaining details, message us on Discord!

Exercise 4: Let $\mathscr{C}$ be a category in which the products $A \times B$ and $C \times D$ exist.

Suppose that the following diagram is a pullback:

$$\begin{tikzcd} R \arrow[r, "e"] \arrow[d, "\langle a,b \rangle"'] & S \arrow[d, "\langle c,d \rangle"] \\ A \times B \arrow[r, "f \times g"] & C \times D \end{tikzcd}$$

Prove that the following diagram is a limiting cone (for a diagram of shape given by the previous exercise):

$$\begin{tikzcd} A \arrow[d, "f"'] & R \arrow[l, "a"'] \arrow[r, "b"] \arrow[d, "e"] & B \arrow[d, "g"] \\ C & S \arrow[l, "c"] \arrow[r, "d"'] & D \end{tikzcd}$$

Does the converse hold?